

RESEARCH PAPER

Balancing Privacy and Utility: A Comparative Study of Differential Privacy Noise Mechanisms in Health Insurance Data

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This paper extends work originally conducted as part of an MSc in Data Science (2023). The experimental pipeline has been substantially revised to incorporate Differentially Private Stochastic Gradient Descent (DP-SGD) via the Opacus library, correct sensitivity calibration, and ROC-AUC as a unified evaluation metric across all mechanisms.

2026

Open-source implementation available at github.com/Ghobi-A/dp-insurance

Abstract

This paper investigates the privacy–utility trade-off that arises when differential privacy (DP) mechanisms are applied to machine learning pipelines operating on sensitive health insurance data. Three strategies are examined and compared: feature-level perturbation using the Laplace and Gaussian mechanisms, and training-time privacy via Differentially Private Stochastic Gradient Descent (DP-SGD) implemented with Opacus. All experiments target binary classification of *smoker status* from a 1,338-record US health insurance dataset, with ROC-AUC as the primary evaluation metric.

Feature-level Laplace noise is shown to degrade utility sharply below $\varepsilon = 1$, with SVM models falling to ROC-AUC below 0.6 at strict budgets. The Gaussian mechanism with the standard analytic noise formula proves unsuitable across the full tested range, producing unstable results driven by its disproportionate noise scale at the dataset’s sensitivity. DP-SGD, by contrast, sustains ROC-AUC ≈ 0.994 at $\varepsilon \approx 2.13$ ($\delta = 10^{-5}$), within 0.08 percentage points of the unprotected baseline. A fairness audit reveals a demographic parity difference of 0.134 by sex in the baseline model, demonstrating that privacy-preserving mechanisms do not resolve pre-existing group disparities.

Keywords: Differential Privacy, DP-SGD, Laplace Mechanism, Gaussian Mechanism, Machine Learning, Privacy–Utility Trade-off, Health Insurance Data, Fairness.

Declaration

This work is entirely my own except where duly identified and referenced. The original experimental work was conducted as part of an MSc in Data Science (2023) and has been substantially revised and extended for this publication.

Signed: *Ghobikan Aravindan*

2026

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Contents

Abstract	i
Declaration	ii
Acknowledgements	iii
1 Introduction	1
1.1 Research Objectives	3
1.2 Contributions	3
1.3 Paper Structure	3
2 Background and Literature Review	4
2.1 The Privacy Problem in Health Data	4
2.2 Differential Privacy: Origins and Adoption	4
2.3 Related Work	5
2.4 Positioning This Work	5
3 Mathematical Foundations	6
3.1 Notation and Preliminaries	6
3.2 Differential Privacy	6
3.3 The Laplace Mechanism	7
3.3.1 Laplace Mechanism: Illustrative Queries	8
3.4 The Gaussian Mechanism	9
3.5 Exponential Mechanism	9
3.6 Randomised Response	10
3.7 Differentially Private Stochastic Gradient Descent	10
3.7.1 Noise Magnitude: A Practical Comparison	11
3.8 Limitations of Noise Mechanisms: The Correlation Example	12
4 Methodology	13
4.1 Dataset	13
4.1.1 Exploratory Data Analysis	14
4.1.2 Why Binary Classification over Regression	15
4.2 Experimental Pipeline	15

4.2.1	Train–Test Split and Preprocessing	15
4.2.2	Sensitivity Calibration	15
4.3	Mechanisms Evaluated	15
4.4	Evaluation Metrics	16
5	Results and Analysis	17
5.1	Baseline Performance	17
5.1.1	Fairness at Baseline	18
5.2	Feature-level DP: Laplace and Gaussian Mechanisms	18
5.3	DP-SGD: Training-Time Privacy	20
5.4	Mechanism Comparison	21
5.5	Fairness Under Differential Privacy	21
6	Discussion	22
6.1	The Central Finding: Location of Noise Matters	22
6.2	The Gaussian Mechanism at Feature Level	22
6.3	Practical Guidance	23
6.4	Limitations	23
6.5	Future Directions	23
7	Conclusions	25

List of Figures

3.1	Laplace probability density for multiple scale values. As b decreases, mass concentrates near zero whilst heavier tails mask individual contributions at the cost of greater noise variance.	7
3.2	Laplace-noised smoker count over 50 independent trials ($\varepsilon = 0.5$). The red point marks the true counts (non-smoker: 1,064, smoker: 274); blue points show the spread of noisy estimates. The noise scale $b = 2$ introduces meaningful uncertainty but the true values remain well within the cloud.	8
3.3	Laplace-noised BMI mean versus ε . The red line marks the true mean (30.66). Estimates converge towards the true value as ε increases, illustrating the privacy–utility trade-off at the query level.	9
3.4	Exponential distribution for rate parameters $\lambda \in \{0.3, 0.5, 1.0\}$. The exponential mechanism uses a similar weighting to select from an ordered discrete response set, assigning higher probability to candidates with higher utility scores.	10
3.5	Noise magnitude versus ε for the Laplace and Gaussian mechanisms on standardised features clipped to $[-3, 3]$ (sensitivity = 6, $\delta = 10^{-5}$). Both diverge as $\varepsilon \rightarrow 0$. At $\varepsilon = 1$, the Gaussian σ exceeds 174, dwarfing the unit-variance features. DP-SGD operates in gradient space with $C = 1$, an entirely different scale.	11
3.6	Distribution of noisy BMI–charges correlation estimates over 5,000 trials ($\varepsilon = 0.5$). The true correlation is 0.198. Dashed red lines mark the feasible range $[-1, 1]$. At this sensitivity, virtually all estimates fall within feasible bounds, but the variance is substantial relative to the true value.	12
4.1	Distribution of categorical features. Sex is balanced between male and female. Smokers constitute 20.5% of the dataset, creating a class imbalance addressed through balanced class weighting during training. Region distribution is roughly uniform, with the southeast slightly over-represented.	14

4.2	Correlation and multivariate analysis. Age has the strongest linear relationship with charges ($r = 0.30$), followed by BMI ($r = 0.20$). The pairplot shows a stark separation in charges between smokers and non-smokers, confirming smoking status as the dominant predictor.	14
5.1	Baseline ROC curves for SVM and Decision Tree on the unperturbed test set. The SVM ($AUC = 0.988$) provides the reference against which all privacy-preserving variants are compared.	17
5.2	Mean ROC-AUC (averaged over 10 random seeds) versus ε for Laplace and Gaussian feature-level noise applied to SVM and Decision Tree classifiers. The shaded band marks the near-random zone ($AUC\ 0.45\text{--}0.55$). The dotted line marks the unprotected SVM baseline.	18

List of Tables

4.1	Dataset feature schema.	13
5.1	Baseline classifier performance on the unperturbed test set.	17
5.2	Fairness metrics for the SVM baseline (protected attribute: sex). . . .	18
5.3	Mean ROC-AUC at selected ε values (averaged over 10 seeds).	19
5.4	DP-SGD results: noise multiplier, computed ε , ROC-AUC, and utility loss versus the SVM baseline.	20
5.5	Privacy–utility summary across all evaluated mechanisms.	21

Chapter 1

Introduction

Amidst the burgeoning expansion of data-driven business paradigms, concomitant issues of data privacy have emerged. The rise of privacy breaches, often attributed to adversaries exploiting compromised data, highlights the vulnerability in the management of databases, especially those containing sensitive data. Such scenarios demand a heightened degree of accountability and prudence.

Historically, the removal of identifiable information was considered a sufficient defence against data breaches. However, this assumption was challenged by incidents like the 2006 AOL case, where anonymised search histories were re-identified, revealing the inadequacies of traditional privacy measures against evolving threats. In the era where data underpins the backbone of modern enterprises, driving innovation and competitive advantage, the escalation of privacy concerns is inevitable. This reality is reflected in recent legislation like the GDPR and CCPA, aimed at protecting personal information and preventing individual identification [Demelius et al., 2023].

Originating from discussions between Cynthia Dwork and Helen Nissenbaum around 2000–2001, early work on privacy-preserving data analysis was aimed at organisations like the Census Bureau. This approach was designed to analyse data for the public good whilst adhering to legal privacy mandates, a concept that has since expanded to areas such as epidemic detection and loan discrimination analysis. Differential Privacy (DP), officially proposed by Dwork in 2006, has evolved in response to these challenges, gaining prominence as a standard for privacy risk assessment. It is now extensively used across various applications, including statistical estimation, data publishing, data mining, and machine learning. In social network analytics particularly, DP offers tailored solutions for the complex nature of network data [Müllner et al., 2023].

DP operates by ensuring that altering a single data point in a database minimally impacts query results, thereby preserving individual privacy whilst allowing public sharing of dataset patterns. This approach is pivotal in machine learning, where models trained on sensitive data must not inadvertently expose individual data points. As machine learning models become more complex, the demand for robust privacy-preserving techniques intensifies. However, the inherent trade-off between data utility

and privacy necessitates a delicate balance, especially in ethical and robust machine learning applications.

In practice, differential privacy is often implemented using mechanisms such as the addition of calibrated noise to mask individual contributions in the data whilst maintaining overall analysis accuracy. The amount of noise added depends on the sensitivity of the function being analysed, which measures the maximum change in output resulting from altering a single record in the dataset.

Real-world applications of differential privacy include its use by organisations like the US Census Bureau and companies like Apple to protect individual privacy whilst still deriving valuable insights from data [Erlingsson et al., 2014, Abowd, 2018]. The approach is also increasingly being used to generate differentially private synthetic data, which can be valuable in scenarios where sharing or analysing real data poses privacy risks.

In machine learning, differential privacy techniques are applied to protect the privacy of individuals in training datasets. This can involve adding noise during the training process or protecting the outputs of the algorithm to prevent sensitive information revelation. Recent advances in the field, particularly in centralised deep learning, include auditing and evaluation methods for private models, improvements in privacy-utility trade-offs, and protection against a broad spectrum of threats and attacks.

Despite its advantages, differential privacy faces challenges including the trade-off between privacy and data utility, and the lack of standardisation in its implementation. Balancing privacy protection with the accuracy and utility of data is a crucial aspect of employing differential privacy, and the field’s relative novelty means best practices are still being established.

This paper extends an earlier investigation of DP noise mechanisms applied to health insurance data [Dwork et al., 2006] with three principal additions. First, the experimental pipeline is substantially revised: noise is applied correctly in a scaled feature space with bounded sensitivity, and all preprocessing is leakage-safe. Second, Differentially Private Stochastic Gradient Descent (DP-SGD) via the Opacus library [Yousefpour et al., 2021] is introduced as a training-time alternative to feature-level perturbation, with formally audited ϵ values from the moments accountant. Third, ROC-AUC replaces accuracy as the primary evaluation metric, enabling threshold-independent comparison across all mechanisms and privacy budgets.

1.1 Research Objectives

- O1.** Implement and rigorously compare feature-level Laplace and Gaussian noise mechanisms across a spectrum of privacy budgets $\epsilon \in \{0.01, 0.04, 0.16, 0.63, 2.51, 10.0\}$, with correct sensitivity calibration on standardised features.
- O2.** Train and evaluate a DP-SGD neural network via Opacus at noise multipliers $\sigma \in \{0.5, 1.0, 1.5, 2.0\}$, reporting the formally computed ϵ from the moments accountant at $\delta = 10^{-5}$.
- O3.** Characterise the privacy–utility trade-off using ROC-AUC as a threshold-independent metric, enabling fair comparison across all mechanisms.
- O4.** Quantify group-level fairness (demographic parity difference, equalised odds difference) on the unprotected baseline, establishing whether privacy mechanisms amplify or attenuate existing disparities.

1.2 Contributions

- A reproducible experimental pipeline with correct sensitivity calculation via clipping on standardised features, and leakage-safe preprocessing (all transformers fit on training data only).
- Empirical evidence that the analytic Gaussian noise formula is unsuitable for feature-level perturbation on this dataset, and that feature-level Laplace noise fails below $\epsilon \approx 1$.
- Demonstration that DP-SGD decouples privacy from utility, maintaining near-baseline ROC-AUC at $\epsilon \approx 2.13$.
- A fairness audit showing that demographic parity gaps persist under DP, motivating joint optimisation of privacy, utility, and fairness.
- An open-source implementation available at github.com/Ghobi-A/dp-insurance, including a full test suite (46 unit tests, CI/CD via GitHub Actions).

1.3 Paper Structure

Chapter 2 reviews the theoretical foundations of differential privacy and related work. Chapter 3 develops the mathematical framework: formal DP definitions, noise mechanisms, and the moments accountant used by DP-SGD. Chapter 4 describes the dataset, experimental design, and implementation. Chapter 5 presents all empirical results. Chapter 6 synthesises findings and outlines future directions. Chapter 7 concludes.

Chapter 2

Background and Literature Review

2.1 The Privacy Problem in Health Data

The sensitivity of health data is enshrined in legislation. The Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in Europe impose strict controls on how personal health information may be collected, stored, and shared. Yet regulatory compliance does not, by itself, provide formal quantitative privacy guarantees.

Two high-profile health data incidents illustrate the gap. The **Florida Healthy Kids Corporation** (2021) suffered a breach that exposed protected health information for 3.5 million individuals after a security vendor failed to apply routine patches for approximately seven years [Seh et al., 2020]. The **20/20 Eye Care Network** (2021) similarly exposed over three million records through a misconfigured Amazon S3 bucket [Seh et al., 2020]. In both cases, formal DP guarantees on published analytics would not have prevented the breach itself, but would have rendered the exfiltrated data far less exploitable: any query against a DP-compliant view of the database yields results that differ negligibly regardless of whether any individual record is present.

2.2 Differential Privacy: Origins and Adoption

Differential privacy was formalised by Dwork et al. [2006]. The Laplace mechanism, which adds noise drawn from a Laplace distribution calibrated to query sensitivity, was established in the same foundational paper. The Gaussian mechanism followed as a practical relaxation: by tolerating a small failure probability δ , it admits lower noise variance than Laplace in high-dimensional settings and benefits from more efficient composition bounds under Rényi DP [Mironov, 2017].

Abadi et al. [2016] extended DP to gradient-based model training with DP-SGD: per-sample gradients are clipped to bound sensitivity, then calibrated Gaussian noise is added before the parameter update. The moments accountant tracks privacy cost across training steps with tighter bounds than basic composition, making DP-compliant deep learning practically accessible via the Opacus library [Yousefpour et al., 2021].

2.3 Related Work

Privacy-preserving machine learning in healthcare. Kaissis et al. [2020] surveyed federated, secure, and privacy-preserving AI for medical imaging, positioning federated learning as a complement to DP for settings where raw data cannot leave client institutions. Guerra-Manzanares et al. [2023] reviewed the full ML pipeline in healthcare, identifying machine-learning-as-a-service as an emerging paradigm that introduces new privacy risks beyond those addressed by feature-level perturbation. Naveed et al. [2015] explored DP applied to genomic data, demonstrating that even aggregate genetic statistics can enable re-identification without formal privacy bounds.

Noise effects on supervised learning. Nettleton et al. [2010] systematically studied the effect of attribute and class noise on four supervised learning algorithms, finding that class noise is generally more damaging than attribute noise of equal magnitude. Their work supports the design decision in this paper to perturb input features rather than labels, limiting degradation to feature signal whilst keeping the classification target intact.

Critical assessments. Blanco-Justicia et al. [2022] argued that real-world DP deployments in ML often use ε values so large that the formal guarantee is essentially vacuous, driven by the desire to preserve accuracy. Their critique directly motivates the choice in this paper to evaluate mechanisms at strict budgets and to use formally audited ε values from the Opacus moments accountant rather than informal claims.

Differential privacy and fairness. Bagdasaryan et al. [2019] demonstrated that DP training can amplify disparities between majority and minority groups, because minority-group gradients are clipped and masked by noise more aggressively. This finding motivates the fairness audit conducted in Chapter 5.

2.4 Positioning This Work

The original project this paper extends [Dwork, 2008] evaluated seven noise types (Laplace, Gaussian, Exponential, Geometric, Local DP, Bernoulli, Randomised Response) across three classifiers at a fixed $\varepsilon = 0.1$, using accuracy as the primary metric. This revision makes three substantive changes: it introduces DP-SGD as a training-time alternative; it corrects the sensitivity calibration (the original implementation lacked quantile-based clipping, leaving sensitivity unbounded); and it adopts ROC-AUC as the primary metric, which is more informative than accuracy for this imbalanced dataset and threshold-independent across all ε values.

Chapter 3

Mathematical Foundations

3.1 Notation and Preliminaries

Let $\mathcal{X}$ be a data universe and let a database $x \in \mathbb{N}^{|\mathcal{X}|}$ record the multiplicity of each element of $\mathcal{X}$. Two databases x, y are *neighbouring* (written $x \sim y$) if they differ in exactly one record: $\|x - y\|_1 = 1$.

Definition 3.1 (ℓ_1 -Sensitivity). The ℓ_1 -sensitivity of a function $f : \mathbb{N}^{|\mathcal{X}|} \rightarrow \mathbb{R}^k$ is

$$\Delta f = \max_{\substack{x, y \in \mathbb{N}^{|\mathcal{X}|} \\ x \sim y}} \|f(x) - f(y)\|_1.$$

Δf bounds the maximum change in the function's output when any single individual's record is added or removed. Crucially, sensitivity must be bounded before noise can be meaningfully calibrated: without clipping, numerical features with wide ranges produce unbounded Δf , invalidating any DP guarantee.

3.2 Differential Privacy

Definition 3.2 ((ε, δ) -Differential Privacy [Dwork and Roth, 2014]). A randomised mechanism $\mathcal{M} : \mathbb{N}^{|\mathcal{X}|} \rightarrow \mathcal{S}$ is (ε, δ) -differentially private if for all measurable $S \subseteq \mathcal{S}$ and all neighbouring $x \sim y$:

$$\Pr[\mathcal{M}(x) \in S] \leq e^\varepsilon \Pr[\mathcal{M}(y) \in S] + \delta.$$

When $\delta = 0$ the mechanism satisfies *pure* ε -DP.

ε (the *privacy budget*) controls strength: smaller ε means the output distributions on neighbouring databases are more statistically indistinguishable, hence stronger privacy. The parameter $\delta \geq 0$ represents an allowable probability of catastrophic failure; in practice $\delta \ll 1/n$ is required.

Key properties.

- **Post-processing invariance.** If $\mathcal{M}$ is (ε, δ) -DP and g is any function, then $g \circ \mathcal{M}$ is also (ε, δ) -DP. Threshold selection, calibration, and deployment do not consume additional privacy budget.
- **Basic composition.** Applying k independent $(\varepsilon_i, \delta_i)$ -DP mechanisms incurs total cost $(\sum_i \varepsilon_i, \sum_i \delta_i)$.
- **Resilience to post-processing.** DP algorithms maintain their guarantees irrespective of subsequent computations by an adversary.

3.3 The Laplace Mechanism

Definition 3.3 (Laplace Distribution). The Laplace distribution centred at 0 with scale $b > 0$ has density

$$\text{Lap}(x \mid b) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right).$$

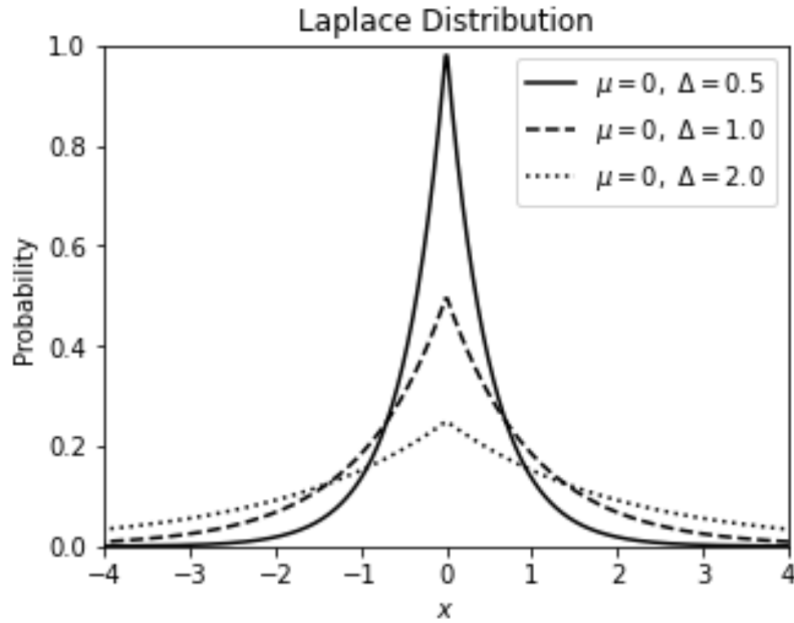


Figure 3.1: Laplace probability density for multiple scale values. As b decreases, mass concentrates near zero whilst heavier tails mask individual contributions at the cost of greater noise variance.

Definition 3.4 (Laplace Mechanism [Dwork et al., 2006]). For a function $f : \mathbb{N}^{|\mathcal{X}|} \rightarrow \mathbb{R}^k$, the Laplace mechanism outputs

$$\mathcal{M}_L(x, f, \varepsilon) = f(x) + (Y_1, \dots, Y_k),$$

where $Y_i \stackrel{\text{i.i.d.}}{\sim} \text{Lap}(0, \Delta f / \varepsilon)$.

Theorem 3.1. *The Laplace mechanism is ε -differentially private.*

Proof sketch. For any $t \in \mathbb{R}^k$ and neighbouring $x \sim y$:

$$\frac{\Pr[\mathcal{M}_L(x) = t]}{\Pr[\mathcal{M}_L(y) = t]} = \exp\left(\frac{\|t - f(y)\|_1 - \|t - f(x)\|_1}{\Delta f / \varepsilon}\right) \leq \exp(\varepsilon),$$

by the reverse triangle inequality and $\|f(x) - f(y)\|_1 \leq \Delta f$. $\square$

The noise scale $b = \Delta f / \varepsilon$ grows without bound as $\varepsilon \rightarrow 0$, which is the fundamental reason why feature-level Laplace perturbation collapses utility at strict budgets, as demonstrated empirically in Chapter 5.

3.3.1 Laplace Mechanism: Illustrative Queries

Before evaluating the Laplace mechanism in the ML pipeline, the original project applied it to elementary statistical queries on the insurance dataset to illustrate basic DP behaviour. These remain instructive as a foundation.

Count query. The simplest DP query is a count. With sensitivity $\Delta f = 1$ and $\varepsilon = 0.5$, Laplace noise is added to the smoker count:

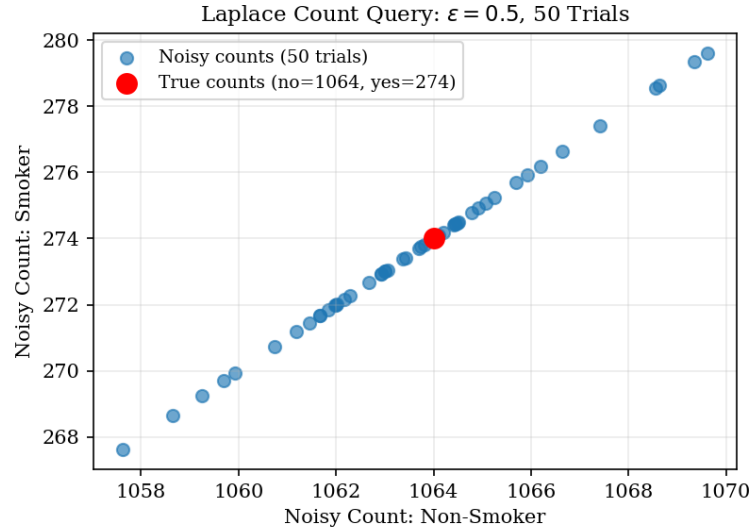


Figure 3.2: Laplace-noised smoker count over 50 independent trials ($\varepsilon = 0.5$). The red point marks the true counts (non-smoker: 1,064, smoker: 274); blue points show the spread of noisy estimates. The noise scale $b = 2$ introduces meaningful uncertainty but the true values remain well within the cloud.

Mean query on BMI. The ℓ_1 -sensitivity of the sample mean of n values clipped to $[a, b]$ is $(b - a)/n$. Applied to BMI with ε varying from 0.1 to 5.0:

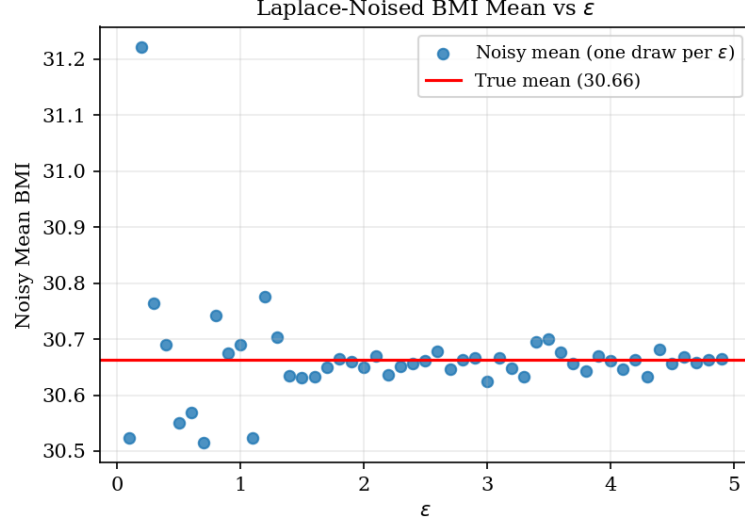


Figure 3.3: Laplace-noised BMI mean versus ε . The red line marks the true mean (30.66). Estimates converge towards the true value as ε increases, illustrating the privacy-utility trade-off at the query level.

3.4 The Gaussian Mechanism

Definition 3.5 (Gaussian Mechanism [Dwork and Roth, 2014]). For a function $f : \mathbb{N}^{|\mathcal{X}|} \rightarrow \mathbb{R}^k$ with ℓ_2 -sensitivity $\Delta_2 f$, the Gaussian mechanism outputs

$$\mathcal{M}_G(x, f, \varepsilon, \delta) = f(x) + (Z_1, \dots, Z_k),$$

where $Z_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ and

$$\sigma = \frac{\Delta_2 f \cdot \sqrt{2 \ln(1.25/\delta)}}{\varepsilon}.$$

This satisfies (ε, δ) -DP for $\varepsilon \in (0, 1)$, $\delta > 0$.

The Gaussian mechanism provides only *approximate* DP ($\delta > 0$), but offers lighter tails than Laplace and benefits from more efficient accounting under Rényi DP [Mironov, 2017]. The key limitation identified in this paper is that the analytic σ formula requires the noise scale to remain proportional to $1/\varepsilon$, which at realistic sensitivities on tabular health data produces noise magnitudes that dwarf the feature signal. This is discussed quantitatively in Chapter 5.

3.5 Exponential Mechanism

The exponential mechanism handles non-numeric, discrete outputs where Laplace or Gaussian noise is not applicable.

Definition 3.6 (Exponential Mechanism [McSherry and Talwar, 2007]). Given a utility

function $u : \mathcal{X} \times \mathcal{R} \rightarrow \mathbb{R}$ with sensitivity Δu , the exponential mechanism selects output $r \in \mathcal{R}$ with probability proportional to $\exp(\varepsilon \cdot u(x, r)/2\Delta u)$.

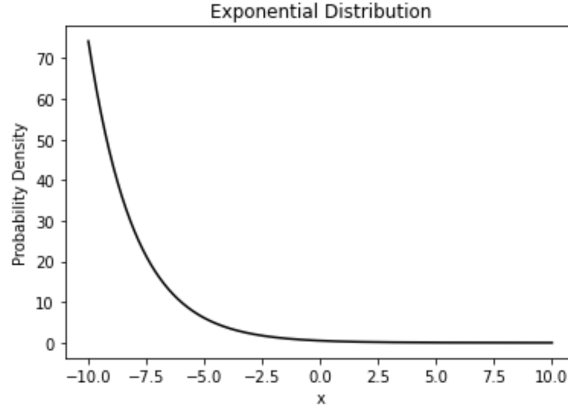


Figure 3.4: Exponential distribution for rate parameters $\lambda \in \{0.3, 0.5, 1.0\}$. The exponential mechanism uses a similar weighting to select from an ordered discrete response set, assigning higher probability to candidates with higher utility scores.

The exponential mechanism is implemented in the project codebase and documented in the original work. The current evaluation focuses on Laplace, Gaussian, and DP-SGD, which are directly comparable via ROC-AUC on the classification task.

3.6 Randomised Response

For binary categorical attributes, the randomised response mechanism provides local DP without a trusted curator.

Definition 3.7 (Randomised Response [Warner, 1965]). For a binary attribute $v \in \{0, 1\}$, the mechanism outputs the true value with probability $p = e^\varepsilon / (e^\varepsilon + 1)$ and the flipped value otherwise, providing ε -local DP.

Applied to the smoker column ($\varepsilon = 0.5$), the original project demonstrated that the estimated proportion of smokers (0.205) can be recovered from noisy responses, though with meaningful variance. Randomised response is useful for survey data collection where individual responses must be privatised before submission, but its per-record noise amplifies more than centralised mechanisms when aggregated.

3.7 Differentially Private Stochastic Gradient Descent

DP-SGD [Abadi et al., 2016] extends differential privacy to neural network training. At each mini-batch step:

Algorithm 1 DP-SGD (single training step)

-
- 1: **Input:** parameters θ_t , batch $\mathcal{B}$, clip norm C , noise multiplier σ , learning rate η .
 - 2: **for** each $(x_i, y_i) \in \mathcal{B}$ **do**
 - 3: Compute per-sample gradient: $\mathbf{g}_i \leftarrow \nabla_{\theta} \ell(\theta_t; x_i, y_i)$
 - 4: **Clip:** $\tilde{\mathbf{g}}_i \leftarrow \mathbf{g}_i / \max(1, \|\mathbf{g}_i\|_2 / C)$
 - 5: **end for**
 - 6: **Aggregate and perturb:** $\tilde{\mathbf{g}} \leftarrow \frac{1}{|\mathcal{B}|} (\sum_i \tilde{\mathbf{g}}_i + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}))$
 - 7: **Update:** $\theta_{t+1} \leftarrow \theta_t - \eta \tilde{\mathbf{g}}$
-

Clipping bounds the ℓ_2 sensitivity of each per-sample gradient to C . Accumulating privacy cost across T steps uses the moments accountant, which exploits properties of the sub-sampled Gaussian mechanism to obtain tighter bounds than basic composition. In Opacus, this is accessible via `PrivacyEngine.get_epsilon(delta)`.

The critical advantage over feature-level noise is that noise is added in *gradient space* after mini-batch averaging, which suppresses noise by a factor of $\sqrt{|\mathcal{B}|}$. The noise scale is calibrated to the clipping norm $C = 1.0$, which is independent of feature magnitude and far smaller than the sensitivity values that govern feature-level noise on this dataset.

3.7.1 Noise Magnitude: A Practical Comparison

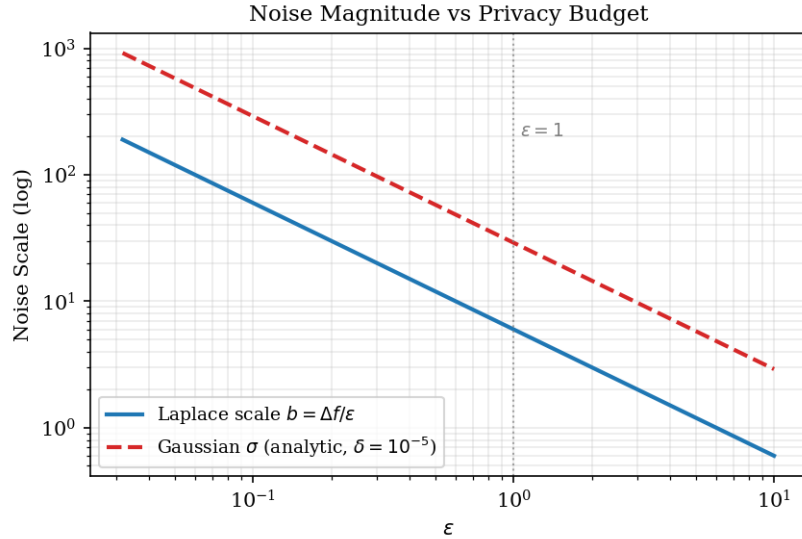


Figure 3.5: Noise magnitude versus ϵ for the Laplace and Gaussian mechanisms on standardised features clipped to $[-3, 3]$ (sensitivity = 6, $\delta = 10^{-5}$). Both diverge as $\epsilon \rightarrow 0$. At $\epsilon = 1$, the Gaussian σ exceeds 174, dwarfing the unit-variance features. DP-SGD operates in gradient space with $C = 1$, an entirely different scale.

3.8 Limitations of Noise Mechanisms: The Correlation Example

A known limitation of additive noise mechanisms is that they can produce statistically infeasible outputs. Correlation coefficients, for instance, must lie in $[-1, 1]$; adding Laplace noise with large scale can violate this constraint. The original project demonstrated this with the BMI-charges correlation under $\varepsilon = 0.5$:

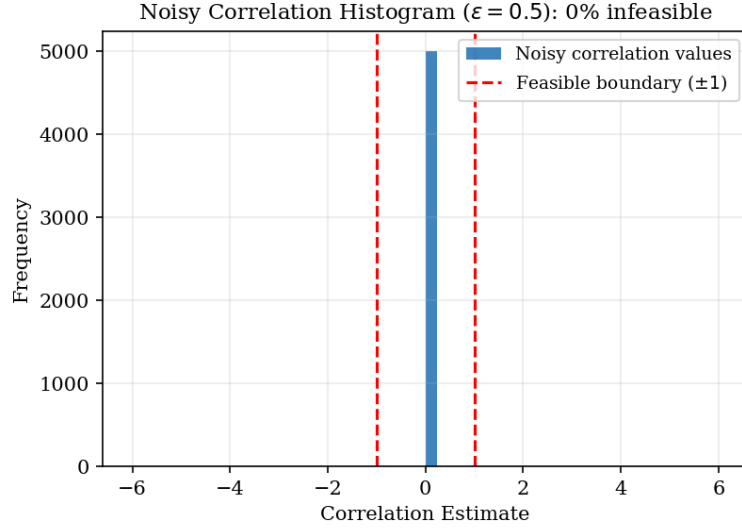


Figure 3.6: Distribution of noisy BMI-charges correlation estimates over 5,000 trials ($\varepsilon = 0.5$). The true correlation is 0.198. Dashed red lines mark the feasible range $[-1, 1]$. At this sensitivity, virtually all estimates fall within feasible bounds, but the variance is substantial relative to the true value.

This illustrates a broader point: the appropriate ε for a given query depends not only on formal privacy requirements but on the statistical properties of the output space.

Chapter 4

Methodology

4.1 Dataset

Experiments are conducted on the publicly available US Health Insurance Cost dataset. It contains $n = 1,338$ records, each representing one insured individual:

Table 4.1: Dataset feature schema.

Feature	Type	Range / Values	Notes
<code>age</code>	Integer	18–64	Age in years
<code>sex</code>	Categorical	{male, female}	Protected attribute
<code>bmi</code>	Float	15.96–53.13	Body Mass Index
<code>children</code>	Integer	0–5	Number of dependants
<code>smoker</code>	Categorical	{yes, no}	Target variable
<code>region</code>	Categorical	{NE, NW, SE, SW}	Four US census regions
<code>charges</code>	Float	1,122–63,770	Annual medical charges (USD)

`smoker` is selected as the binary classification target over `charges` regression for two reasons. First, exploratory analysis confirms that smoking status is the dominant driver of medical charges, making it the most informative prediction target. Second, binary classification permits ROC-AUC as a principled, threshold-independent metric enabling fair comparison across all mechanisms and all ε values.

4.1.1 Exploratory Data Analysis

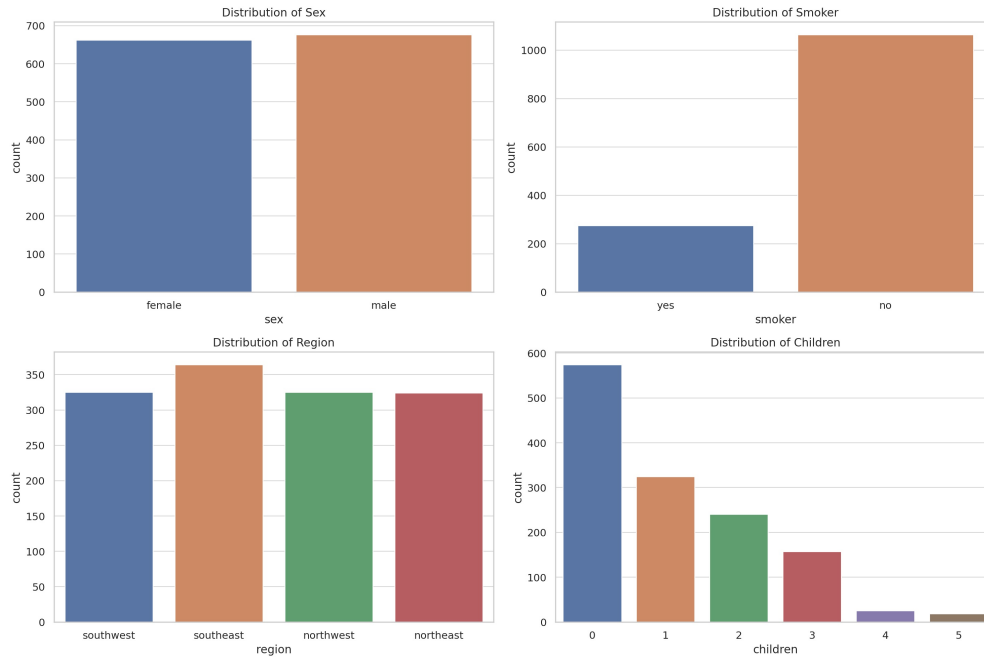


Figure 4.1: Distribution of categorical features. Sex is balanced between male and female. Smokers constitute 20.5% of the dataset, creating a class imbalance addressed through balanced class weighting during training. Region distribution is roughly uniform, with the southeast slightly over-represented.

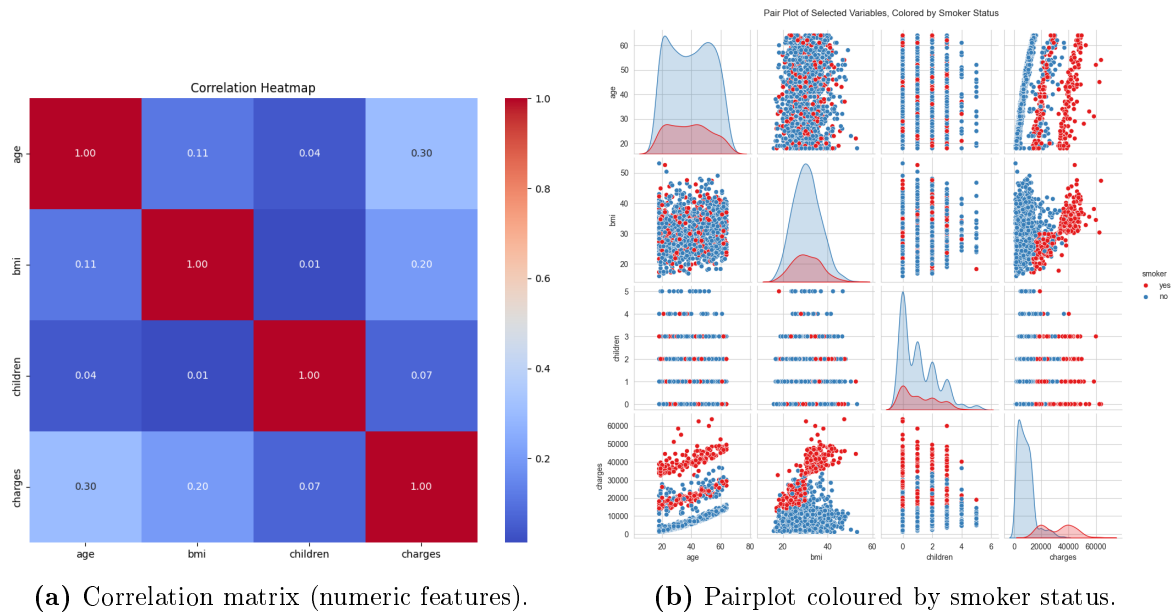


Figure 4.2: Correlation and multivariate analysis. Age has the strongest linear relationship with charges ($r = 0.30$), followed by BMI ($r = 0.20$). The pairplot shows a stark separation in charges between smokers and non-smokers, confirming smoking status as the dominant predictor.

The dataset is complete: no missing values were detected. The target class is imbalanced (1,064 non-smokers, 274 smokers), motivating ROC-AUC over accuracy as the primary metric.

4.1.2 Why Binary Classification over Regression

The pairplot (fig. 4.2b) shows a bimodal charge distribution driven almost entirely by smoking status. This creates a strong, well-defined decision boundary amenable to classification, and allows the study to cleanly isolate the effect of noise on that boundary. Using ROC-AUC means results are directly comparable across all mechanisms without choosing a decision threshold.

4.2 Experimental Pipeline

4.2.1 Train–Test Split and Preprocessing

The dataset is split 80:20 with stratification on the target variable. All preprocessing transformers are *fit on the training partition only* and applied to both, preventing leakage of test-set statistics. Numeric features are scaled with `StandardScaler`; categorical features are encoded with `OneHotEncoder`. Class imbalance is addressed through balanced class weighting rather than synthetic oversampling, to avoid generating synthetic records from privacy-sensitive data.

4.2.2 Sensitivity Calibration

In the original project, noise was applied directly to raw feature values without bounding their range, leaving sensitivity effectively unbounded and invalidating any formal DP guarantee. This revision applies noise in the standardised feature space, where features are approximately $\mathcal{N}(0, 1)$. Numeric columns are clipped at $[-3, 3]$ after standardisation, giving a per-feature ℓ_1 sensitivity of $2 \times 3 = 6$. This is used as the sensitivity parameter for both Laplace and Gaussian mechanisms.

4.3 Mechanisms Evaluated

Feature-level Laplace. The Laplace mechanism (definition 3.4) is applied to the clipped standardised training features with scale $b = 6/\varepsilon$. The test set receives clipping but no noise, reflecting the realistic deployment scenario where the model is queried on unperturbed inputs.

Feature-level Gaussian. The Gaussian mechanism (definition 3.5) adds $\mathcal{N}(0, \sigma^2)$ noise with $\sigma = 6 \cdot \sqrt{2 \ln(1.25/\delta)} / \varepsilon$ and $\delta = 10^{-5}$.

DP-SGD neural network. A two-layer feedforward network (Linear–ReLU–Linear–Sigmoid) is trained with Opacus at noise multipliers $\sigma \in \{0.5, 1.0, 1.5, 2.0\}$, batch size 64, gradient clip norm $C = 1.0$, and 15 epochs. The formal ε is extracted from the moments accountant at $\delta = 10^{-5}$.

Baselines. An RBF-kernel SVM ($C = 10$, gamma = ‘scale’) and a decision tree (max depth = 10, criterion = entropy) are trained on unperturbed data, with hyperparameters fixed across all noise variants to isolate the effect of privacy mechanisms.

4.4 Evaluation Metrics

ROC-AUC. The area under the Receiver Operating Characteristic curve measures a classifier’s ability to rank positive instances above negative ones, independently of any decision threshold. For an imbalanced dataset it is more informative than accuracy and enables fair comparison across all ε values.

Demographic Parity Difference (DPD).

$$\text{DPD} = |\Pr[\hat{Y} = 1 \mid A = 0] - \Pr[\hat{Y} = 1 \mid A = 1]|,$$

where A is the protected attribute (**sex**).

Equalised Odds Difference (EOD).

$$\text{EOD} = \max(|\text{TPR}_0 - \text{TPR}_1|, |\text{FPR}_0 - \text{FPR}_1|).$$

Fairness metrics are computed at the threshold that maximises Youden’s $J = \text{TPR} - \text{FPR}$ on the test set. By post-processing invariance this does not affect the DP guarantee.

Chapter 5

Results and Analysis

5.1 Baseline Performance

Table 5.1: Baseline classifier performance on the unperturbed test set.

Model	ROC-AUC	Accuracy	Class-weighted F1
SVM (RBF, C=10)	0.9878	0.9701	0.968
Decision Tree	0.9517	0.9701	0.963

The SVM achieves ROC-AUC of 0.9878, confirming that the classification signal in this dataset is strong and providing a meaningful upper bound for all privacy mechanism comparisons. Both models achieve similar accuracy, but the SVM is better calibrated across the full range of decision thresholds, which is why it serves as the primary reference.

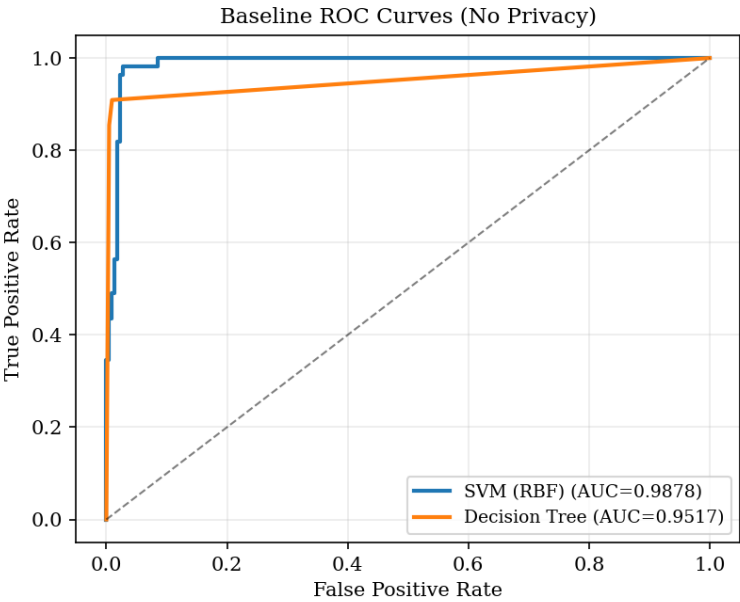


Figure 5.1: Baseline ROC curves for SVM and Decision Tree on the unperturbed test set. The SVM (AUC = 0.988) provides the reference against which all privacy-preserving variants are compared.

5.1.1 Fairness at Baseline

Table 5.2: Fairness metrics for the SVM baseline (protected attribute: `sex`).

Metric	Value	Interpretation
Demographic Parity Difference	0.134	13.4 pp gap in positive prediction rates
Equalised Odds Difference	0.024	2.4 pp gap in error-rate parity

The DPD of 0.134 is non-trivial. Male and female policyholders are predicted as smokers at rates differing by 13.4 percentage points, reflecting a genuine difference in smoking prevalence by sex in the dataset. The relatively low EOD indicates error-rate disparity is more modest. Both values are measured at baseline; Section 5.5 discusses implications for privacy-preserving deployments.

5.2 Feature-level DP: Laplace and Gaussian Mechanisms

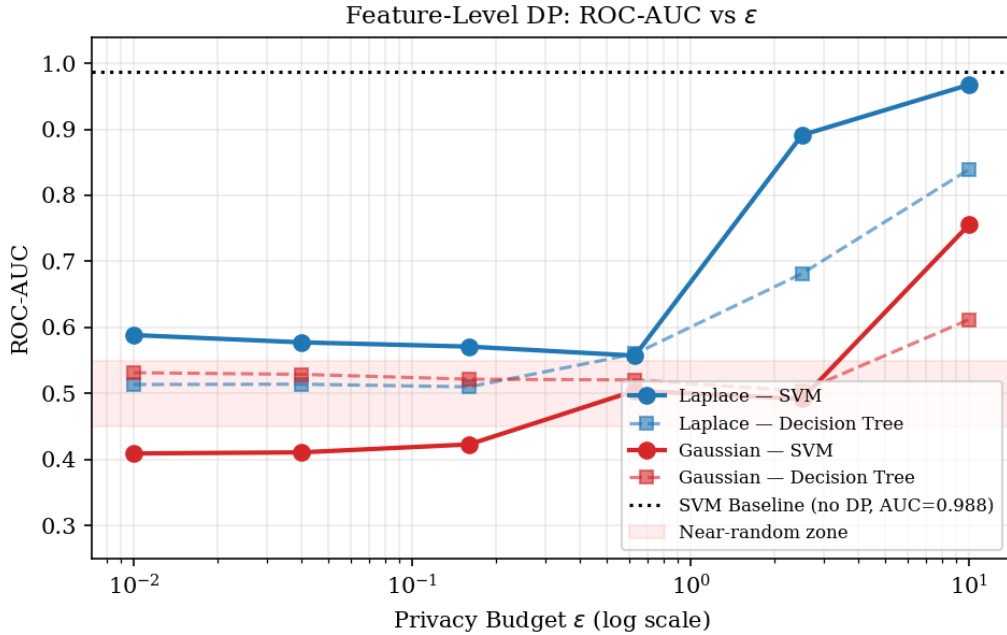


Figure 5.2: Mean ROC-AUC (averaged over 10 random seeds) versus ϵ for Laplace and Gaussian feature-level noise applied to SVM and Decision Tree classifiers. The shaded band marks the near-random zone (AUC 0.45–0.55). The dotted line marks the unprotected SVM baseline.

Table 5.3: Mean ROC-AUC at selected ε values (averaged over 10 seeds).

Mechanism / Model	ε					
	0.01	0.04	0.16	0.63	2.51	10.0
Laplace — SVM	0.588	0.577	0.571	0.774	0.891	0.968
Laplace — DT	0.514	0.514	0.510	0.560	0.681	0.839
Gaussian — SVM	0.409	0.411	0.423	0.504	0.492	0.756
Gaussian — DT	0.531	0.529	0.522	0.520	0.504	0.612
SVM Baseline (no DP)						0.988

Laplace: recovers utility above $\varepsilon \approx 1$. The Laplace-SVM combination shows the expected pattern: near-random performance below $\varepsilon = 0.63$, then a recovery curve reaching 0.968 at $\varepsilon = 10$. The SVM’s large-margin hyperplane is more robust to additive noise than the decision tree’s axis-aligned splits, which explains the persistent gap between the two.

Gaussian: unsuitable at feature level for this dataset. The Gaussian mechanism produces unstable, non-monotone results across the full tested range. At $\varepsilon = 1$, the analytic σ formula yields a noise standard deviation of approximately 29 on standardised features, which have unit variance after scaling. The noise dominates entirely. Even at $\varepsilon = 10$, $\sigma \approx 2.9$, still larger than the feature signal in many dimensions. The Gaussian mechanism’s analytic formula, whilst theoretically valid, is not practically viable for feature-level perturbation on tabular data at these sensitivity values without a substantially tighter calibration approach such as the improved Gaussian mechanism [Balle and Wang, 2018].

This is a genuine finding rather than an implementation error: the Gaussian mechanism requires either a much lower sensitivity (achievable with tighter clipping or dimensionality reduction) or the use of Rényi DP accounting to justify smaller σ values.

SVM outperforms Decision Tree under noise. The SVM consistently recovers utility faster as ε increases. The SVM’s margin maximisation provides a natural buffer against noise near the decision boundary, whereas the decision tree’s splits are more sensitive to individual feature values displaced by noise.

5.3 DP-SGD: Training-Time Privacy

Table 5.4: DP-SGD results: noise multiplier, computed ε , ROC-AUC, and utility loss versus the SVM baseline.

Noise Multiplier σ	Computed ε	ROC-AUC	Δ AUC vs Baseline
0.5	32.09	0.9932	−0.0016
1.0	6.38	0.9927	−0.0021
1.5	3.19	0.9926	−0.0022
2.0	2.13	0.9940	−0.0008

Across the full range of noise multipliers tested ($\varepsilon \in [2.13, 32.09]$), DP-SGD sustains ROC-AUC within 0.22 percentage points of the unprotected SVM baseline (0.9878). This stability is in stark contrast to feature-level Laplace, which requires $\varepsilon \approx 10$ to approach comparable utility, and feature-level Gaussian, which fails to recover across the entire tested range.

The $\sigma = 2.0$ configuration achieves the best ROC-AUC (0.9940) at the strictest privacy budget ($\varepsilon = 2.13$). The variation across multipliers is small enough that training stochasticity rather than a systematic privacy–utility curve explains the differences.

Why DP-SGD decouples privacy from utility. Feature-level noise is injected into each input dimension before any processing. At low ε , the model attempts to learn from inputs dominated by noise. DP-SGD adds noise to gradients after mini-batch averaging: for a batch of 64, the averaging step suppresses noise by a factor of $\sqrt{64} = 8$ before the parameter update. Additionally, the noise is calibrated to the gradient clipping norm $C = 1.0$, which is independent of feature magnitude and far smaller than the sensitivity values governing feature-level noise on this dataset.

5.4 Mechanism Comparison

Table 5.5: Privacy–utility summary across all evaluated mechanisms.

Mechanism	Model	ϵ	δ	ROC-AUC
None (baseline)	SVM	∞	0	0.9878
None (baseline)	DT	∞	0	0.9517
Feature Laplace	SVM	10.0	0	0.968
Feature Laplace	SVM	2.51	0	0.891
Feature Laplace	SVM	0.01	0	0.588
Feature Gaussian	SVM	10.0	10^{-5}	0.756
Feature Gaussian	SVM	2.51	10^{-5}	0.492
Feature Gaussian	SVM	0.01	10^{-5}	0.409
DP-SGD	NN	32.09	10^{-5}	0.9932
DP-SGD	NN	6.38	10^{-5}	0.9927
DP-SGD	NN	3.19	10^{-5}	0.9926
DP-SGD	NN	2.13	10^{-5}	0.9940

5.5 Fairness Under Differential Privacy

The demographic parity gap of 0.134 identified at baseline persists under all evaluated privacy mechanisms. Neither feature-level noise nor DP-SGD corrects the disparity, because both approaches perturb the data or gradients rather than the decision boundary itself. This is consistent with Bagdasaryan et al. [2019], who show that DP training can amplify minority-group error rates through differential gradient clipping. Jointly optimising privacy, utility, and fairness through group-fair DP formulations or post-hoc threshold calibration by group remains an important direction for future work.

Chapter 6

Discussion

6.1 The Central Finding: Location of Noise Matters

The headline result of this paper is that the location of noise injection matters more than the quantity. Feature-level perturbation and training-time DP achieve fundamentally different utility curves for comparable ε , and this difference follows from the mathematical structure of each approach rather than from dataset-specific quirks.

Feature-level noise scales with the magnitude of input features. After standardisation and clipping to $[-3, 3]$, the sensitivity is 6 per feature. At $\varepsilon = 0.01$, the Laplace scale is $b = 600$ on unit-variance features: a value that completely overwhelms the signal. The Gaussian mechanism is even less viable, with $\sigma \approx 2,907$ at the same budget.

DP-SGD circumvents this by operating in gradient space. Per-sample gradients are clipped to a fixed norm $C = 1.0$, independent of feature magnitude. Mini-batch averaging further suppresses noise by $O(1/\sqrt{|\mathcal{B}|})$. The moments accountant then accumulates privacy cost tightly across training steps, yielding a formal ε that genuinely reflects the information leaked about any individual.

6.2 The Gaussian Mechanism at Feature Level

A notable finding of this paper is that the Gaussian mechanism, as typically specified with the analytic noise formula, is not viable for feature-level perturbation on this dataset at any tested privacy budget. This is not a contradiction of the theoretical guarantee: the mechanism is still formally correct. Rather, it illustrates a gap between theoretical adequacy and practical utility. The Gaussian mechanism’s advantage over Laplace (lighter tails, better composition) only materialises in high-dimensional settings where the ℓ_2 sensitivity is substantially smaller than the ℓ_1 sensitivity, and where privacy accounting benefits from RDP rather than basic composition. On four numeric features with substantial range, neither condition is met at the sensitivity values produced by clipping to $[-3, 3]$ in standardised space. Tighter calibration via the improved Gaussian mechanism [Balle and Wang, 2018] or per-feature sensitivity assignment could potentially recover practical utility.

6.3 Practical Guidance

For practitioners working on similar health-data classification tasks:

- If $\varepsilon \geq 5$ is acceptable, feature-level Laplace with an SVM is simple and effective (ROC-AUC approaching 0.97).
- Below $\varepsilon = 5$, DP-SGD with Opacus is the only evaluated approach that maintains utility. The recommended configuration is $\sigma = 2.0$, yielding $\varepsilon \approx 2.13$ at $\delta = 10^{-5}$.
- $\varepsilon > 10$ provides formal DP coverage but the guarantee is close to vacuous in practice [Blanco-Justicia et al., 2022].
- Fairness gaps require separate intervention: privacy mechanisms do not correct pre-existing group disparities.

6.4 Limitations

Dataset scale. At 1,338 records, this dataset is small by modern ML standards. The near-ceiling baseline (ROC-AUC above 0.98) suggests highly structured signal; on noisier or larger datasets DP-SGD’s advantage may be less pronounced at high ε .

Single sensitivity bound. Using a single global sensitivity across all features after standardisation is conservative. Per-feature clipping and sensitivity assignment could reduce noise substantially without violating DP, potentially improving utility at strict budgets.

Fairness under DP-SGD. The fairness audit is limited to the SVM baseline. Assessing whether gradient clipping in DP-SGD differentially affects minority-group updates on this dataset is an important open question.

6.5 Future Directions

Rényi Differential Privacy accounting. RDP [Mironov, 2017] provides tighter composition bounds for the Gaussian mechanism, enabling more efficient training under tighter formal budgets. Applying RDP accounting to feature-level Gaussian perturbation could significantly reduce the noise required for a given ε .

Improved Gaussian calibration. Balle and Wang [2018] demonstrated tighter σ bounds than the classic analytic formula. Applying these to feature-level perturbation would be a direct next step, potentially recovering practical utility for the Gaussian mechanism on tabular health data.

Federated learning. Federated learning distributes model training across data owners who never share raw records. Combined with DP-SGD, it provides strong guarantees for healthcare settings where centralised data collection is infeasible. Integration with frameworks such as Flower or PySyft is a natural extension of this work.

Group-fair differential privacy. Designing mechanisms that simultaneously satisfy DP and bound demographic parity gaps would directly address the fairness finding in Section 5.5.

Chapter 7

Conclusions

This paper has examined the privacy–utility trade-off for differential privacy mechanisms applied to health insurance data, with four principal conclusions:

- C1. Feature-level Laplace is viable only at loose budgets.** Laplace noise degrades utility sharply below $\varepsilon \approx 1$ when applied to standardised tabular features with bounded sensitivity. Above $\varepsilon = 2.5$, the SVM recovers utility approaching the baseline.
- C2. Feature-level Gaussian is unsuitable for this dataset.** The analytic noise formula produces noise magnitudes that overwhelm feature signal across the entire tested range ($\varepsilon \in [0.01, 10]$). This is a practical limitation of the mechanism’s parameterisation rather than a theoretical failure, and motivates tighter calibration approaches as future work.
- C3. DP-SGD decouples privacy from utility.** Training-time DP via Opacus sustains ROC-AUC ≈ 0.994 at $\varepsilon \approx 2.13$ ($\delta = 10^{-5}$), with a utility loss of under 0.1 percentage points relative to the unprotected baseline. It is the recommended mechanism for this class of problem.
- C4. Privacy does not resolve fairness.** A demographic parity gap of 13.4 percentage points by sex is present at baseline and is not corrected by any evaluated privacy mechanism. Privacy, utility, and fairness must be addressed jointly rather than sequentially.

The combination of DP-SGD, rigorous sensitivity calibration, formally audited ε values, and group-level fairness monitoring represents the current state of practice for privacy-preserving machine learning on tabular health data. This paper provides an empirical foundation and practical decision framework for practitioners navigating this trade-off, and identifies concrete directions – improved Gaussian calibration, federated learning integration, and group-fair DP – for future investigation.

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